

CGPA CALCULATOR USING C++

This project calculates the CGPA of a student based on grades and credit hours entered by the user. The program uses vectors, loops, and conditional statements in C++.

Program Code:

```
#include <iostream>
#include <vector>
#include <iomanip>
using namespace std;

int main() {
    int n;
    cout << "Enter number of courses: ";
    cin >> n;

    vector<string> grades(n);
    vector<int> credits(n);

    float totalCredits = 0;
    float totalPoints = 0;

    for (int i = 0; i < n; i++) {
        cout << "\nCourse " << i + 1 << endl;

        cout << "Enter grade (A/B/C/D/F): ";
        cin >> grades[i];

        cout << "Enter credit hours: ";
        cin >> credits[i];

        int point;

        if (grades[i] == "A")
            point = 10;
        else if (grades[i] == "B")
            point = 8;
        else if (grades[i] == "C")
            point = 6;
        else if (grades[i] == "D")
            point = 4;
        else
            point = 0;

        totalCredits += credits[i];
        totalPoints += point * credits[i];
    }

    float cgpa = totalPoints / totalCredits;

    cout << fixed << setprecision(2);
    cout << "\nFinal CGPA: " << cgpa << endl;

    return 0;
}
```

Sample Output:

Output

Enter number of courses: 5

Course 1

Enter grade (A/B/C/D/F): A

Enter credit hours: 4

Course 2

Enter grade (A/B/C/D/F): A

Enter credit hours: 4

Course 3

Enter grade (A/B/C/D/F): A

Enter credit hours: 4

Course 4

Enter grade (A/B/C/D/F): B

Enter credit hours: 3

Course 5

Enter grade (A/B/C/D/F): A

Enter credit hours: 4

Final CGPA: 9.68